CS301: Database Management Systems - Sample Notes

Unit 3: Normalization and Functional Dependencies

1NF: Every attribute must contain atomic values.

2NF: 1NF plus no partial dependency on composite candidate keys.

3NF: 2NF plus no transitive dependencies on candidate keys.

BCNF: For every functional dependency X -> Y, X must be a superkey.

Unit 4: Transaction Management

ACID Properties: Atomicity, Consistency, Isolation, Durability.

Two-Phase Locking guarantees conflict serializability.